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A hybrid stock price index forecasting model based on variational mode decomposition and LSTM network

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which the original complex series can be decomposed into a limited number of subseries with relatively simple modes of fluctuations. It can effectively overcome the shortcomings of mode mixing that sometimes exist in the empirical mode decomposition (EMD) method. LSTM is an improved version of recurrent neural networks (RNNs) that introduces a “gate” mechanism, and can effectively filter out the critical previous information, making it suitable for the financial time series forecasting. The capability of VMD-LSTM in stock price index forecasting is verified comprehensively by comparing with some single models and the EMD-based and other VMD-based hybrid models. Evaluated by level and directional prediction criteria, as well as a newly introduced statistic called the complexity-invariant distance (CID), the VMD-LSTM model shows an outstanding performance in stock price index forecasting. The hybrid models perform significantly better than the single models, and the forecasting accuracy of the VMD-based models is generally higher than that of the EMD-based models.

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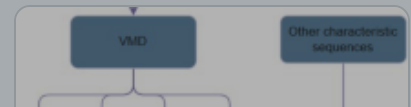
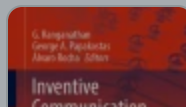
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